The Hall Effect

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Introduction:

The Hall effect occurs when a material carrying a current through it is placed within a perpendicular magnetic field, causing the moving charged carrier particles to be pushed towards one side of the field by experiencing a magnetic force. The resulting separation of positive and negative charges creates an electric field around the current-carrying material, which leads to observing a measurable electric potential difference known as Hall voltage. The objective of this experiment is to apply this principle by placing a Germanium wafer semiconductor (current-carrying material) in a perpendicular magnetic field using two permanent magnets. By observing the Hall voltage generated by the current and magnetic field, the experiment aims to determine if the semiconductor is of p-type or n-type, calculate the carrier density, and measure the charge carrier mobility within the wafer.

Theory:

In semiconductors like germanium, each atom forms covalent bonds with each other using four valence electrons. At room temperature, thermal energy breaks a small amount of these bonds producing a free electron which is a mobile negative charge. A hole is formed from the lack of electrons to bond with which acts as a mobile positive charge as neighboring bonded electrons may break from their current bonds to bond with the hole. In pure germanium, there is only a small amount of free electrons and holes and therefore the conductivity is low. However, atoms of either three or five valence electrons can be added in a process called doping. By adding five valence electron atoms, they bond with the germanium and produce a lot of free electrons with the extra valence electrons which becomes the majority carrier and turns the germanium into an n-type semiconductor. By adding three-valence electron atoms, they bond with the germanium and produce a lot of holes with the lack of valence electrons. These become the majority carrier turning the germanium into a p-type semiconductor. By putting a current (represented by $I = ne|v_d|Wt$ where n is the charge carrier density, e is the charge of each carrier, v_d is the drift velocity, W is the width of the wafer and t is the thickness of the wafer) through the length of a doped germanium wafer, free electrons flow against the direction of conventional current while holes flow with the direction of conventional current. By putting a magnetic field perpendicular to the direction of current, the moving carrier experiences a magnetic force of $F_b = qv_d \times B$ where q is the charge and B is the magnetic field. Since both the sign of charge and direction of drift velocity is reversed for free electrons, both carrier types will experience a force in the same direction pushing them to one edge of the wafer. As carriers accumulate on one side of the wafer, it creates a net charge which leaves an opposite charge on the other side. The combination of both creates a transverse electric field called the Hall field E_H which exerts a force opposite to the direction of the magnetic force. This process will continue until enough carriers pile up on one side to where the electric and magnetic forces are equal. At this point, carriers will drift straight through as if no magnetic field was placed and a potential difference called the Hall voltage V_H now exists along the width of the wafer where $|\Delta V_h| = W|v_d||B|$. By combining this with the current formula, you get:

$$|\Delta V_h| = W|v_d||B| \qquad (1)$$

$$I = ne|v_d|Wt \qquad (2)$$

$$rearrange\ (2)\ \Rightarrow |v_d| = \frac{I}{neWt}\ \Rightarrow\ sub\ into\ (1)\ \Rightarrow |\Delta V_h| = W\frac{I}{neWt}|B|\ \Rightarrow\ |\Delta V_h| = \frac{B}{n\cdot e\cdot t}I$$

$$|\Delta V_h| = \frac{B}{n\cdot e\cdot t}I \qquad (3)$$

This means, for a set magnetic field, the Hall voltage should have a linear relationship with current and the slope of the relationship between these two will yield the carrier density n. The carrier mobility $\mu = \frac{|v_d|}{|E_L|}$ describes how readily carriers drift in the longitudinal electric field $E_L = \frac{\Delta V_L}{L}$. By combining these with equation (1), you get:

$$|\Delta V_h| = W|v_d||B| \qquad (1)$$

$$\mu = \frac{|v_d|}{|E_L|} \qquad (5)$$

$$E_L = \frac{\Delta V_L}{L} \qquad (6)$$

$$Substitute\ (6)\ into\ (5) \Rightarrow \mu = \frac{|v_d|}{\frac{\Delta V_L}{L}} \Rightarrow |v_d| = \frac{\mu \Delta V_L}{L}$$

$$rearranging\ for\ |v_d|\ \Rightarrow |v_d| = \frac{\mu \Delta V_L}{L} \qquad (7)$$

$$substitute\ (7)\ into\ (1)\ \Rightarrow |\Delta V_h| = W\frac{\mu \Delta V_L}{L}|B|$$

$$final\ equation\ \Rightarrow |\Delta V_h| = (\frac{\mu WB}{L})\Delta V_L \qquad (8)$$

This shows that ΔV_h has a linear relationship with ΔV_L. By comparing the Hall voltage with the known directions of current and magnetic field along with the voltmeter connects, we can identify the wafer as either n-type or p-type.

Procedure:

The tip of the Gauss meter was first placed inside the zero-gauss chamber, and the "push to zero" knob was pushed to zero the Gauss meter. The probe was then inserted between the permanent magnets with the blue lettering facing upwards to measure the magnetic field strength of the magnets. 5 total measurements of the magnetic field were recorded, with each measurement being taken from a different area of the permanent magnet to ensure an accurate reading was recorded. For this experiment, a reading of 0.2198 T was recorded to determine the strength of the magnetic field. The power supply was then connected to the germanium wafer board with the positive terminal of the power supply connected to the positive end of the wafer, and the negative terminal connected to ground. The current on the power supply was adjusted to 50 mA. The multimeter was then connected to the germanium wafer with the +V terminal connected to the positive terminal of the board and the COM terminal connected to ground. To measure the correct Hall voltage, the multimeter was connected to terminals on the board closer to the centre, as those ports measure Hall voltage as shown in Figure 1. The zero knob on the semiconductor board was then adjusted until a reading of approximately 0 Volts was observed on the multimeter. The Germanium wafer was then inserted between the magnetic poles, and the board was centred by lining up the lines on the wafer with the scratch on the plexiglass. The general direction of the current, magnetic field, and

multimeter connections were noted down. The current was varied starting from 50 mA and decreasing by increments of 5 mA until the lowest reading was observed on the power supply. The power supply provided within the lab had a lowest current reading of 1.4 mA. At each of these current measurements, the Hall voltage was recorded on a table with the use of the multimeter. The multimeter wires were then taken out of the terminals on the board and placed into the same ports as the DC power supply to measure the voltage across the length of the wafer as shown in Figure 2. The DC power supply was then set to 50mA and decreased by increments of 5 mA until the lowest reading on the DC power supply was observed. At each of these new currents, the longitudinal voltage was observed and recorded with the use of a multimeter. Finally, the power supply was then turned off, and the multimeter was disconnected from the Germanium wafer.

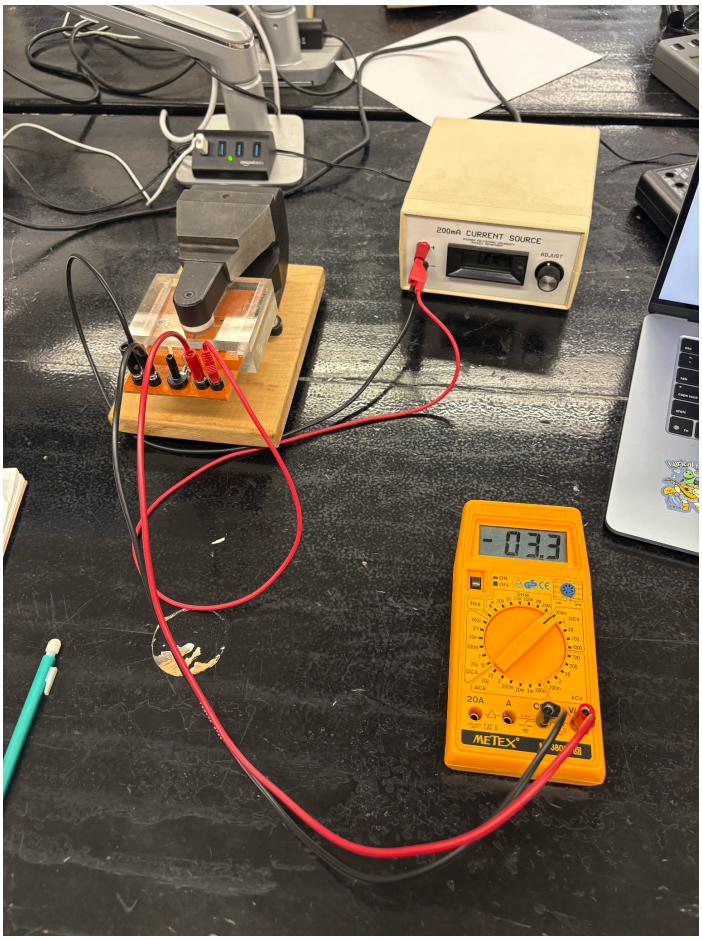


Figure 1: Measuring the Hall Voltage Using the Multimeter

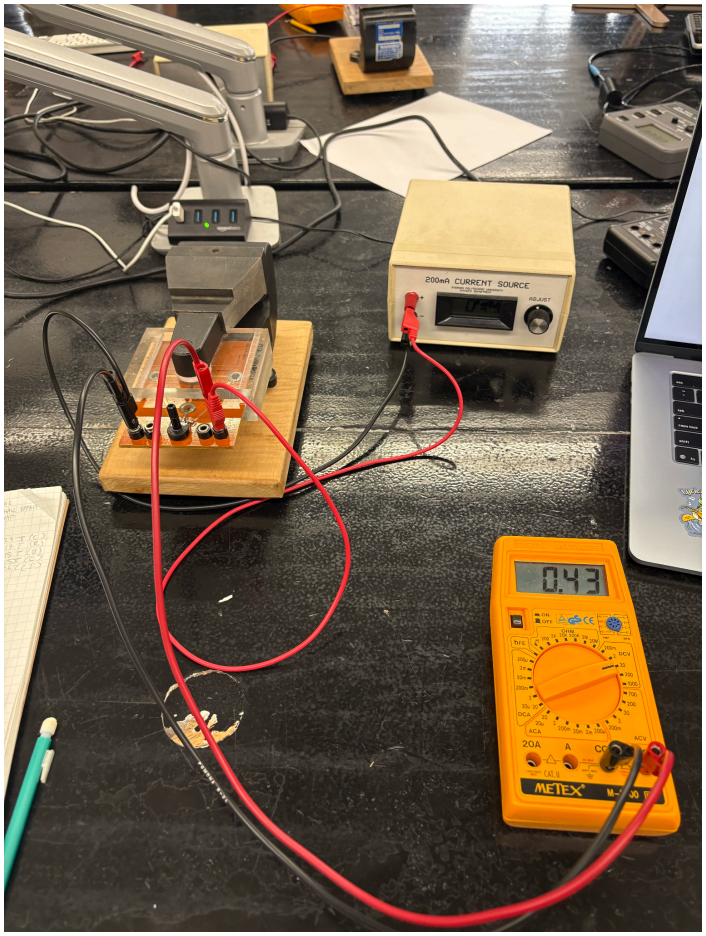


Figure 2: Measuring the Longitudinal Voltage Using the Multimeter

Results and Calculations:

Table 1: Recorded Observations For The Magnetic Field Generated By The Permanent Magnet and Dimensions of the Germanium Wafer:

Parameter	Value	Uncertainty	Units
Magnetic Field ($\lvert\vec{B}\rvert$)	0.2198	$\pm$ 0.0005	Tesla (T)
Length (L)	10.0	n/a	mm
Width (W)	5.0	n/a	mm
Thickness (t)	1.0	n/a	mm

From the Gauss meter, a positive reading was obtained for the magnetic field measurement ($\vec{B}$). Since the probe was inserted into the permanent magnetic field with the blue lettering facing upwards, following the standard convention of the Gauss meter tells us that the magnetic field generated by the permanent magnet is pointing in the downward direction. To ensure that an accurate reading was taken to

determine the magnetic field, 5 different observations were recorded using the Gauss meter, with each reading taken at a different position of the permanent magnet. The average of these values was taken to determine the final result of the magnetic field. The calculations to find the average magnetic field strength are as follows:

Let B represent the different readings taken on the permanent magnet

$$B_{center} = 0.223\,T\,;\ B_{right} = 0.217\,T\,;\ B_{left} = 0.218\,T\,;\ B_{bottom} = 0.215\,T\,;\ B_{top} = 0.226\,T$$

$$B_{average} = |\vec{B}| = \frac{0.223 + 0.217 + 0.218 + 0.215 + 0.226}{5}\,T = 0.2198\,T$$

Therefore, the average strength of the magnetic field generated by the permanent magnet is $0.2198\,T$.

1) Using the sign of the Hall voltage which you observed, and comparing your experimental setup with your work on the pre-lab questions, determine whether this germanium wafer is n-type or p-type.

Table 2: Recorded Observations of the Current Supplied By The Source (I), Hall Voltage (ΔV_H), and Voltage Across the Length Of The Germanium Wafer (ΔV_L):

Current $I\ (mA)$	**Uncertainty** $\delta I\ (mA)$	**Hall Voltage** $\Delta V_H\ (mV)$	**Uncertainty** $\delta V_H\ (mV)$	**Longitudinal Voltage** $\Delta V_L\ (V)$	**Uncertainty** $\delta V_L\ (V)$
50.0	$\pm\ 0.05$	-122.7	$\pm\ 0.05$	4.83	$\pm\ 0.005$
45.0	$\pm\ 0.05$	-110.8	$\pm\ 0.05$	4.34	$\pm\ 0.005$
40.0	$\pm\ 0.05$	-98.7	$\pm\ 0.05$	3.85	$\pm\ 0.005$
35.0	$\pm\ 0.05$	-86.5	$\pm\ 0.05$	3.36	$\pm\ 0.005$
30.0	$\pm\ 0.05$	-74.2	$\pm\ 0.05$	2.87	$\pm\ 0.005$
25.0	$\pm\ 0.05$	-61.7	$\pm\ 0.05$	2.38	$\pm\ 0.005$
20.0	$\pm\ 0.05$	-49.4	$\pm\ 0.05$	1.90	$\pm\ 0.005$
15.0	$\pm\ 0.05$	-36.9	$\pm\ 0.05$	1.41	$\pm\ 0.005$
10.0	$\pm\ 0.05$	-24.5	$\pm\ 0.05$	0.94	$\pm\ 0.005$
5.0	$\pm\ 0.05$	-12.3	$\pm\ 0.05$	0.46	$\pm\ 0.005$
1.4	$\pm\ 0.05$	-3.6	$\pm\ 0.05$	0.12	$\pm\ 0.005$

The experimental results show that the germanium wafer is an n-type semiconductor. Based on the data found in Table 2, the measured Hall Voltage (ΔV_H) consistently reads a value of less than 0 Volts, spanning from $-\ 122.7\ mV$ to $-\ 3.6\ mV$. In our setup, the conventional current provided by the power supply flows towards the left while the magnetic field points downwards onto the Germanium Wafer. By applying the right-hand rule, the charged particles get pushed towards the front of the wafer. Because the Hall Voltage reading was negative, a build-up of negatively charged particles (electrons) was observed at the positive terminal of the multimeter. Therefore, since the experimental Hall Voltage readings were less than 0 Volts, the active charge carriers are electrons, confirming that the Germanium wafer semiconductor is n-type

2) Plot your data of ΔVH and I in such a way so that the result is a straight line. Add error bars to your plot. Fit the data with a line, and use your answers to the pre-lab questions to determine the carrier concentration n from the slope. Make sure to check the units!

From the Hall voltage equation (1) and current equation (2), a linear relationship between the two variables can be derived as follows:

$$|\Delta V_h| = W|v_d||B| \qquad (1)$$

$$I = ne|v_d|Wt \qquad (2)$$

$$rearrange\ (2) \Rightarrow |v_d| = \frac{I}{neWt} \Rightarrow sub\ into\ (1) \Rightarrow |\Delta V_h| = W\frac{I}{neWt}|B| \Rightarrow |\Delta V_h| = \frac{B}{n \cdot e \cdot t}I$$

$$|\Delta V_h| = \frac{B}{n \cdot e \cdot t}I \qquad (3)$$

Therefore, the linear relationship between the Hall Voltage and Current can be described by equation (3), plotted as shown in Figure 3.

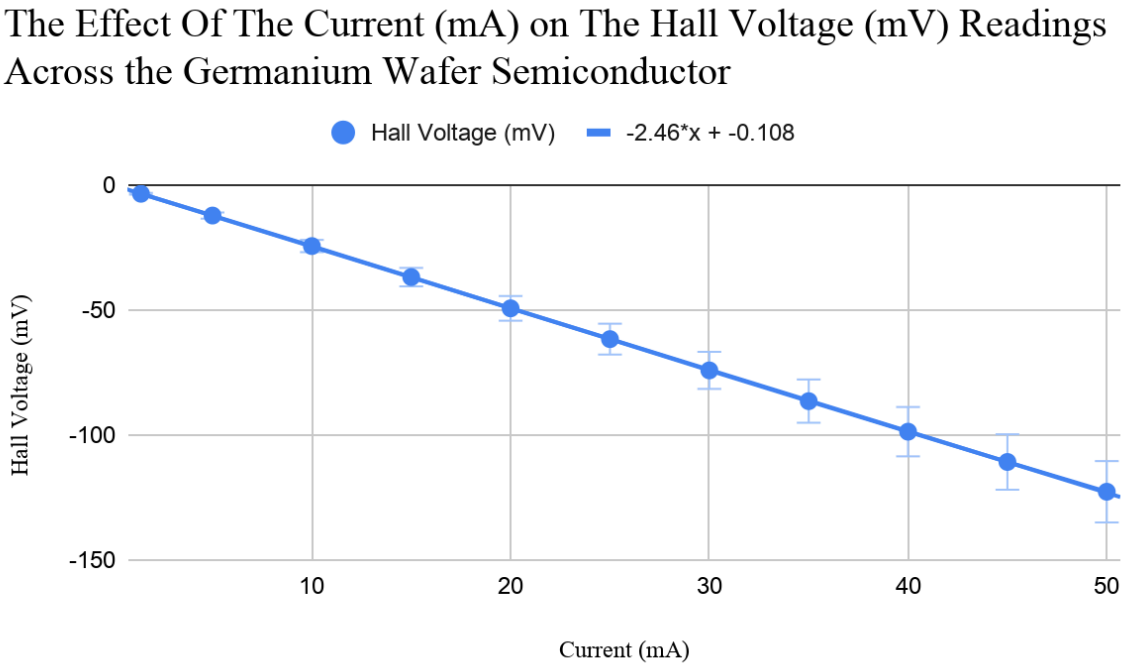


Figure 3: The Effect Of The Current (mA) on the Hall Voltage (mV) Readings Across the Germanium Wafer Semiconductor: From this graph, the carrier concentration, n, can be determined using the slope ($m = -2.46$) of the graph.

From the slope of the graph, the carrier concentration can be calculated as shown below:

Givens: $m = -2.46$; $|\vec{B}| = 0.2198\,T$; $e = -1.602 \times 10^{-19} C$; $t = 1.0 \times 10^{-3}\,m$

$$m = \frac{B}{n \cdot e \cdot t} \qquad (4)$$

Rearranging equation (4) for carrier concentration n

$$n = \frac{B}{m \cdot e \cdot t} = \frac{0.2198\,T}{(-2.46)(-1.602\times10^{-19} C)(1.0\times10^{-3}\,m)} = 5.577 \times 10^{20}\,m^{-3}$$

Therefore, the carrier concentration, n, is approximately $5.577 \times 10^{20}\,m^{-3}$.

3) Plot your data of ΔVH and ΔVL in such a way so that the result is a straight line. Add error bars to your plot. Fit the data with a line, and use your answers to the pre-lab questions to determine the carrier mobility μ from the slope. Make sure to check the units!

From the carrier mobility equation (5), the Hall Voltage equation (1) and the electric field equation (6), a linear relationship between the Hall Voltage and longitudinal voltage can be described as:

$$|\Delta V_h| = W|v_d||B| \quad (1)$$

$$\mu = \frac{|v_d|}{|E_L|} \quad (5)$$

$$E_L = \frac{\Delta V_L}{L} \quad (6)$$

$$Substitute\ (6)\ into\ (5) \Rightarrow \mu = \frac{|v_d|}{\frac{\Delta V_L}{L}} \Rightarrow |v_d| = \frac{\mu \Delta V_L}{L}$$

$$rearranging\ for\ |v_d| \Rightarrow |v_d| = \frac{\mu \Delta V_L}{L} \quad (7)$$

$$substitute\ (7)\ into\ (1) \Rightarrow |\Delta V_h| = W\frac{\mu \Delta V_L}{L}|B|$$

$$final\ equation \Rightarrow |\Delta V_h| = (\frac{\mu WB}{L})\Delta V_L \quad (8)$$

Therefore, the linear relationship between the Hall Voltage and longitudinal voltage can be described by equation (8), plotted as shown in Figure 4.

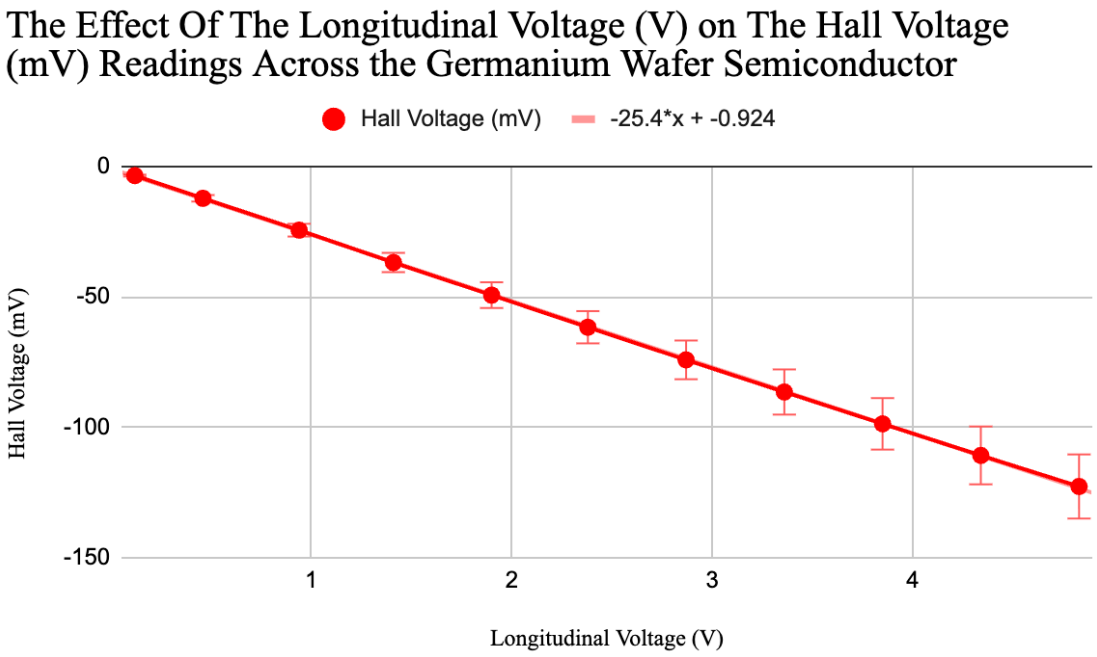


Figure 4: The Effect Of The Longitudinal Voltage (V) on the Hall Voltage (mV) Readings Across the Germanium Wafer Semiconductor: From this graph, the carrier mobility, μ, can be determined using the slope ($m = -25.4\ mV/V = -0.0254\ V/V$) of the graph.

From the slope of the graph, the carrier mobility can be calculated as shown below:

Givens: $m = -0.0254$; $|\vec{B}| = 0.2198\ T$; $W = 5.0 \times 10^{-3} m$; $L = 10.0 \times 10^{-3}\ m$

$$m = \frac{\mu WB}{L} \quad (9)$$

Rearranging (9) for the carrier mobility

$$\mu = \left|\frac{mL}{WB}\right| = \left|\frac{0.0254(10.0\times10^{-3}\,m)}{(5.0\times10^{-3}m)(0.2198T)}\right| = 0.2311\,\frac{m^2}{V\cdot s} = 2311\,\frac{cm^2}{V\cdot s}$$

Therefore, the carrier mobility, μ, is approximately $2311\,\frac{cm^2}{V\cdot s}$.

4) Compare your value of the mobility μ to the accepted value for germanium, which has μn = 3900 cm2/V · s and μp = 1900 cm2/V · s. Calculate the % error from the accepted value.

The experimentally obtained carrier mobility, μ, is approximately $2311\,\frac{cm^2}{V\cdot s}$. The theoretical value of the carrier mobility is $\mu = 3900\,\frac{cm^2}{V\cdot s}$. This shows that there are discrepancies between the value obtained and what was expected, pointing towards uncertainty and experimental error. The percentage error can be determined as shown here:

$$\%\,Error = \left|\frac{Experiment - Theory}{Theory}\right| \times 100\% = \left|\frac{2311-3900}{3900}\right| \times 100\% = 40.7\%$$

Therefore, the percent error between the carrier mobilities experimental and the accepted value is approximately 40.7%.

Discussion and Conclusions:

Both plots displayed clear linear relationships as predicted by equations (3) and (8). For the Hall voltage vs current, each decrease of 5 mA in current decreased the Hall voltage by a near constant amount of about 12.3 mV with the fitted line passing very close to the origin (intercept ≈ -0.1mV). This confirms that Hall voltage is directly proportional to current. Similarly, the ratio of Hall voltage to longitudinal voltage remained approximately constant at about −25 mV/V across all data points, showing that Hall voltage is also directly proportional to longitudinal voltage. The Hall voltage was negative at every current, indicating that electrons accumulated on the edge connected to the positive terminal of the multimeter. Comparing this with the known directions of the current and magnetic field confirmed that the germanium wafer is an n-type semiconductor. The measured carrier density, $n \approx 5.58 \times 10^{20}\,m^{-3}$ is roughly 23 times the intrinsic carrier density value of germanium at room temperature ($n_i \approx 2.4 \times 10^{19} m^{-3}$). This is consistent with a doped semiconductor where the donor atoms provide much more free electrons compared to thermal generation alone and supports our n-type identification. The measured mobility of 2311 cm²/V·s is 40.7% below the accepted electron mobility of 3900 cm²/V·s. The statistical uncertainty in the fitted slope is only about 0.3%, and the spatial variation in the magnetic field readings is about ±2.5%. Therefore such large error percentage likely comes from sources like contact resistance from the contacts and leads, which adds to the measured ΔV_L and therefore lowers the calculated μ. Heat energy from the electrical energy of the current increases lattice vibrations which lowers μ and magnetic field non-uniformity which can directly affect both n and μ.

The Hall effect was used to characterize a germanium semiconductor wafer. The consistently negative Hall voltage showed that the wafer is n-type, with electrons as the majority carriers. The linear relationship between Hall voltage and current gave us a carrier density of 5.58×10^{20} m^{-3} which is consistent with a doped semiconductor. The relationship between Hall voltage and longitudinal voltage gave an electron mobility of 2311 $cm^2/V \cdot s$. This is 40.7% below the accepted value of 3900 $cm^2/V \cdot s$, attributed mainly to contact resistance, Joule heating of the wafer and magnetic field non-uniformity.

References:

Honsberg, C., & Bowden, S. (n.d.). *Equilibrium carrier concentration*. PVEducation. https://www.pveducation.org/pvcdrom/pn-junctions/equilibrium-carrier-concentration

Scientific Equipment & Services. (n.d.). *User's manual: Hall effect setup*. https://www.iiserkol.ac.in/~ph324/ExptManuals/HallEffect.pdf

University of Rhode Island. (n.d.). *Semiconductor properties* [ELE 533 course material]. https://www.ele.uri.edu/Courses/ele533/05/midterm/Physicaldata.html

Appendix:

1. No, the sign of ΔV_H only tells you the carrier type relative to the orientation of the current, magnetic field and voltmeter leads. If you flip any of these, it'll flip the sign of the voltage. Even if you have a fixed setup, the statement fails when you have both carrier types present in comparable amounts in the semiconductor. Doping produces the carriers while thermal energy creates free electrons and holes. The concentrations of the two are defined by the law of mass action $n_0 p_0 = n_i^2$. When current is added, both carriers pile up on the same edge as magnetic force is in the same direction for both while the drift velocity is opposite with opposite charges so the net force for both carriers is in the same direction. Since both carriers are on the same edge with opposite charges, they partially cancel each other's Hall voltage. The remaining Hall coefficient is modeled by $R = \frac{\mu_h^2 p - \mu_e^2 n}{e(\mu_h p + \mu_e n)^2}$ where μ_h is the mobility of holes, μ_e is the mobility of electrons, p is the density of holes, n is the density of electrons and e is the elementary charge. The sign of R/V_H is set by the numerator as the denominator can never be negative. Therefore the sign of the Hall effect is determined by which carrier dominates the Hall effect when weighted by mobility squared and not just which carrier is majority. When you compared the two, you get $\frac{\mu_n^2}{\mu_p^2} = (\frac{3900\ cm^2/V \cdot s}{1900 cm^2/V \cdot s})^2 \approx 4.2$ therefore each electron affects about 4.2 times as much as each hole. So the holes have to outnumber the electrons by over 4 times in order to get a positive Hall voltage and a p type semiconductor with a 3 to 1 ratio of holes to electrons can have a negative Hall voltage. In our experiment, the orientation of the current, magnetic field and meter leads was recorded and compared with the pre-lab figures. In addition, the measured carrier density ($n \approx 5.6 \times 10^{20}\ m^{-3}$) is much larger than the intrinsic carrier density of germanium ($n_i \approx 2.4 \times 10^{19}\ m^{-3}$). By the law of mass action, the hole density is approximately

$p = n_i^2/n \approx 1 \times 10^{18}\, m^{-3}$, roughly 550 electrons for every hole. Since electrons dominate by far more than the 4:1 weighting, the negative Hall voltage reliably indicates that the wafer is n-type.

2. The conductivity of the wafer is expected to be larger than that of intrinsic germanium as conductivity depends on both the number of charge carriers and their mobility ($\sigma = en\mu$). Intrinsic germanium has relatively few carriers as its produced only by thermal generation ($n_i \approx 2.4 \times 10^{19}\, m^{-3}$),while doping adds a large number of free electrons from donor atoms. Since the measured carrier density ($5.58 \times 10^{20}\, m^{-3}$) is approximately 23 times the intrinsic value, a significantly higher conductivity is expected. Using the measured values, we get:
$\sigma = en\mu = (1.602 \times 10^{-19}\, C)(5.578 \times 10^{20}\, m^{-3})(0.2311\, m^2/V{\cdot}s) \approx 20.7\ (\Omega{\cdot}m)^{-1}$

   This is approximately 9.4 times the intrinsic value of $2.2\ (\Omega{\cdot}m)^{-1}$, confirming the expectation. The increase is smaller than the 23-fold increase in carrier density because intrinsic germanium conducts using both electrons and holes, while the doped wafer conducts primarily through electrons, and because the measured mobility was lower than the accepted value.

3. If the magnitude of the magnetic field were increased while all other parameters remained the same, the magnitude of the Hall voltage would increase. From equation (3), since the current, carrier density, elementary charge and wafer thickness are all unchanged, the Hall voltage is directly proportional to the magnetic field strength. A stronger magnetic field exerts a larger magnetic force on each drifting carrier. To reach equilibrium, the Hall field must exert an equal and opposite force, so a larger Hall field is required. This requires more charge to accumulate on the edges of the wafer, producing a larger potential difference across its width. The slope of the Hall voltage versus current graph would increase, while the carrier density calculated from it would remain unchanged.
4. If the magnetic field were increased while everything else remained the same, the calculated mobility would remain the same. The mobility was determined from the slope of the Hall voltage versus longitudinal voltage graph using $\mu = \frac{mL}{WB}$, where $m = \frac{\mu WB}{L}$. Increasing the magnetic field increases the Hall voltage proportionally, while the longitudinal voltage is unchanged because it only depends on the current and the resistance of the wafer. The slope therefore increases in proportion to B, but since B also appears in the denominator of the mobility equation, the two effects cancel. For example, doubling the field to 0.4396 T would double the slope to $0.0508\ \frac{V}{V}$, giving $\mu = \frac{(0.0508)(0.010\ m)}{(0.005\ m)(0.4396\ T)} = 0.2311 \frac{m^2}{V{\cdot}s}$, identical to the original result..